

Project Presentation

MedGuardAI

Intelligent Multi-Agent System for Medicine Authentication and Safety Verification

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OUTLINE

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INTRODUCTION & PROBLEM STATEMENT

- Growing issue of counterfeit, expired, and illegal medicines
- Difficult for users to verify medicine authenticity in real-time
- Existing systems lack multi-modal verification and accessibility
- Need for an AI-based, user-friendly verification solution

Problem Statement : *Counterfeit and expired medicines pose serious health risks, while existing barcode-based systems are unreliable due to damage, duplication, and lack of real-time verification. There is a need for a simple, integrated system using AI, OCR, and barcode technology to ensure accurate and accessible medicine authentication.*



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OBJECTIVES

1. Develop an AI-based medicine verification system
2. Enable verification using text, image (OCR), and barcode
3. Classify medicines as:
 - a. Authentic
 - b. Fake
 - c. Expired
 - d. Unverified
4. Implement similarity search using pgvector (RAG)
5. Provide safe symptom-based guidance using local LLM



CONCEPT & METHODS

Concept : MedGuard AI is an AI-powered medicine verification system that detects counterfeit, expired, or unsafe drugs. It analyzes images, text, and barcodes to provide real-time, reliable, and user-friendly authentication for consumers and healthcare professionals.

Methods :

1. Multi-Modal Input Processing – Handles text, image, and barcode inputs.
2. OCR (Google Cloud Vision API) – Extracts text from medicine packaging.
3. Barcode Decoding – Identifies product information.
4. Information Extraction & Validation – Processes and verifies data.
5. Semantic Search (RAG + pgvector/FAISS) – Retrieves relevant medicine data.
6. Verification Logic – Checks authenticity and expiry.
7. LLM-Based Summarization (AI Models) – Generates simple results.
8. Microservice Architecture – Ensures scalability and modular design

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LITERATURE REVIEW

Ref	Paper	Contribution	Outcome	Limitation
[1]	Motwani et al., 2022 -Counterfeit MedicineDetection using DeepLearning	Used CNN-based imagerecognition and OCR forcounterfeit detection.	Achieved ~90% accuracy inidentifying real vs fakepackaging.	Limited dataset size; lacksbarcode handling and real-world variability.
[2]	Renushe et al., 2024 - A QRCode Approach ToCounterfeit Drug Prevention	Developed a QR/Barcode-based medicineauthentication system.	Enabled quick verificationthrough QR scanning andserialization.	Fails if code is duplicatedor damaged.
[3]	Liu et al., 2020 - DLI-IT:Drug Label Identificationthrough Image and TextEmbedding	Proposed hybrid modelcombining image featuresand OCR text embeddings.	Improved drug labelrecognition accuracy to88%.	Computationally intensive;limited generalization forunseen data.
[4]	Naughton et al., 2016 -Effectiveness of MedicinesAuthentication Technology	Analyzed serialization-based authentication inhospital systems.	Detected counterfeit andexpired drugs effectively inreal conditions.	Dependent on manualscanning and infrastructurequality.



IDENTIFICATION OF GAPS & SCOPE OF WORK

Gaps :

1. Existing systems rely on single-mode verification and lack a unified multi-modal approach (MedScan, Zydus Verify)
2. Many solutions require costly infrastructure (blockchain, IoT, lab setup), limiting real-world use (ITS Mobile, Blockchain Drug Trace Systems)
3. Current methods are not scalable or user-friendly for general consumers (PharmaSecure psVerify)
4. Lack of advanced AI integration and poor handling of real-world issues (damaged packaging, duplicate barcodes) (QR Code Verification Systems, MedScan)

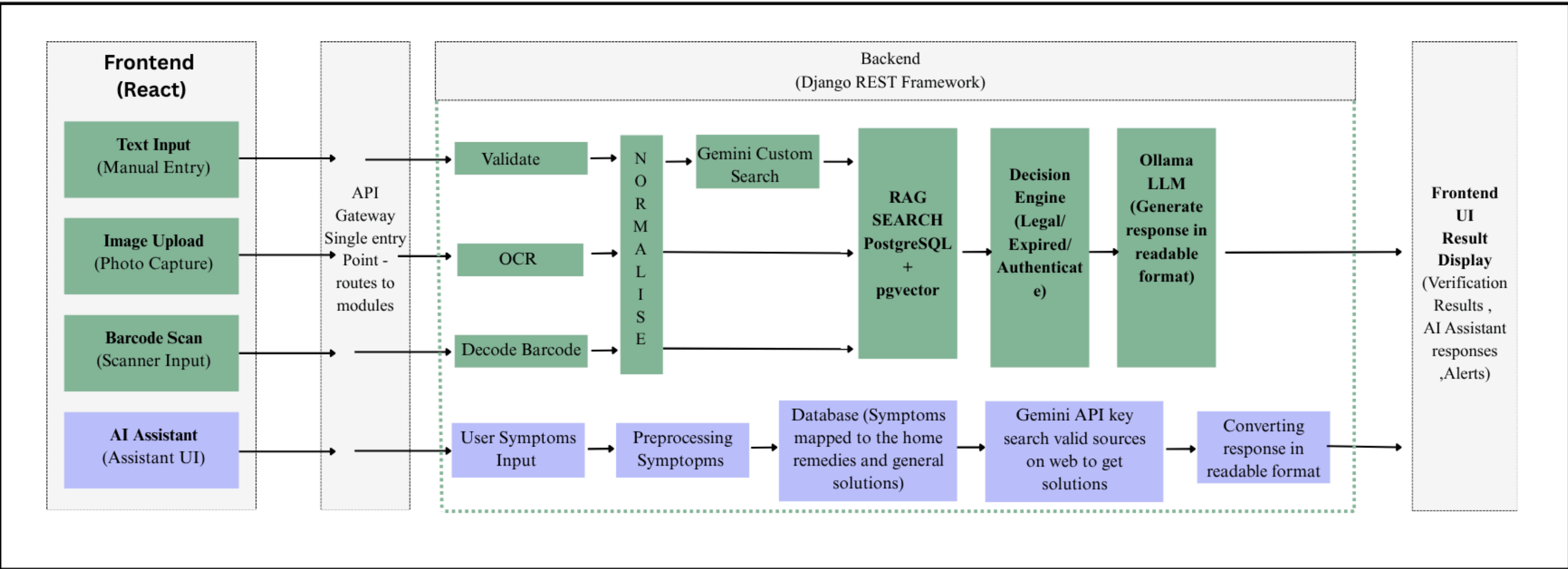
Scope of Work :

1. Build a complete multi-modal verification system
2. Improve accuracy using AI + similarity search
3. Develop scalable backend using Django REST Framework
4. Enable real-time verification and user history tracking
5. Expand dataset and improve model performance



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PROCESS AND ARCHITECTURE



TOOLS AND TECHNOLOGIES

1. Backend: Django REST Framework
2. Database: PostgreSQL
3. RAG: pgvector
4. OCR: PaddleOCR, Tesseract
5. LLM: Ollama (local)
6. Frontend: React, Flutter
7. Authentication: JWT
8. Deployment: Docker, Cloud



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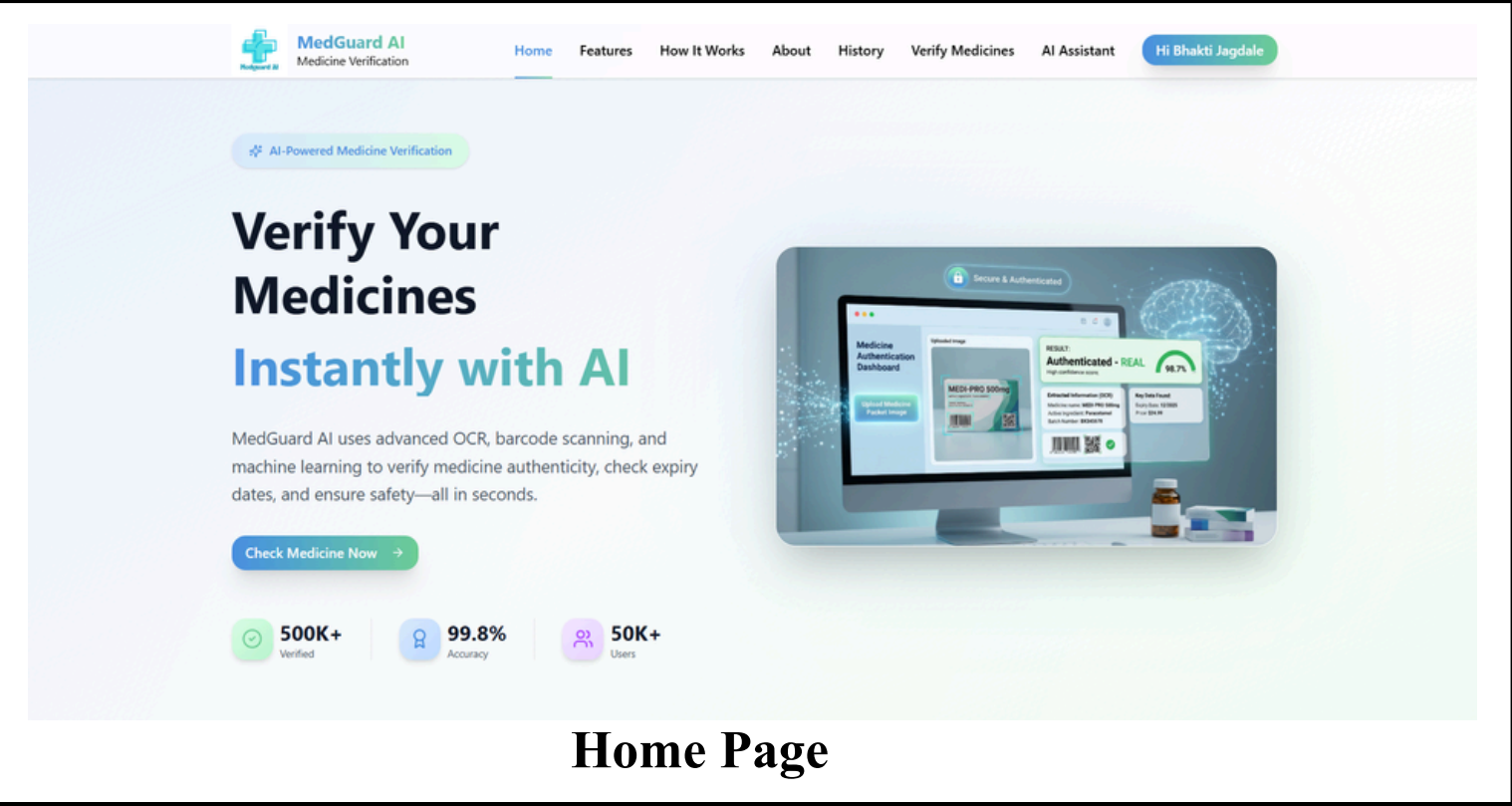
IMPLEMENTATION

- Developed API services using Django REST Framework
- Integrated OCR for image-based text extraction
- Implemented barcode scanning and decoding
- Built classification logic for medicine verification
- Stored data and embeddings using PostgreSQL + pgvector
- Connected frontend with backend APIs
- Containerized services using Docker

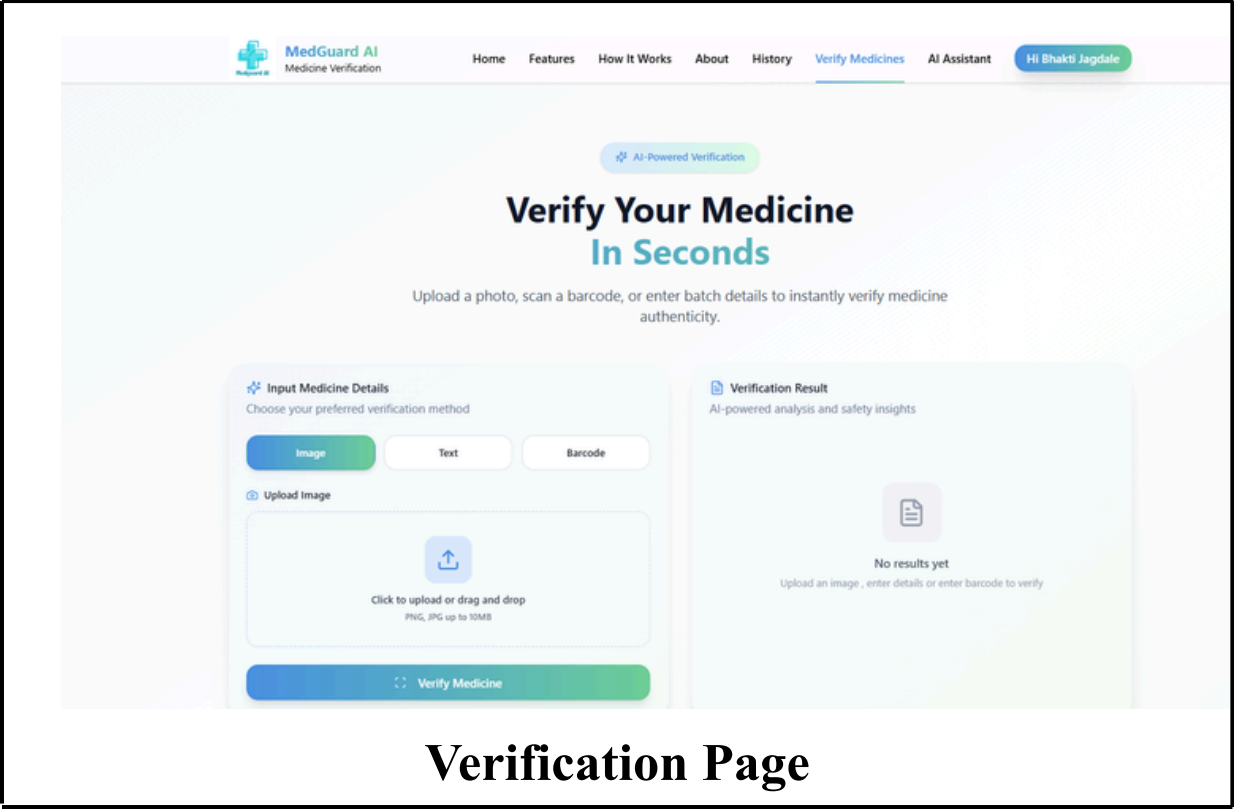


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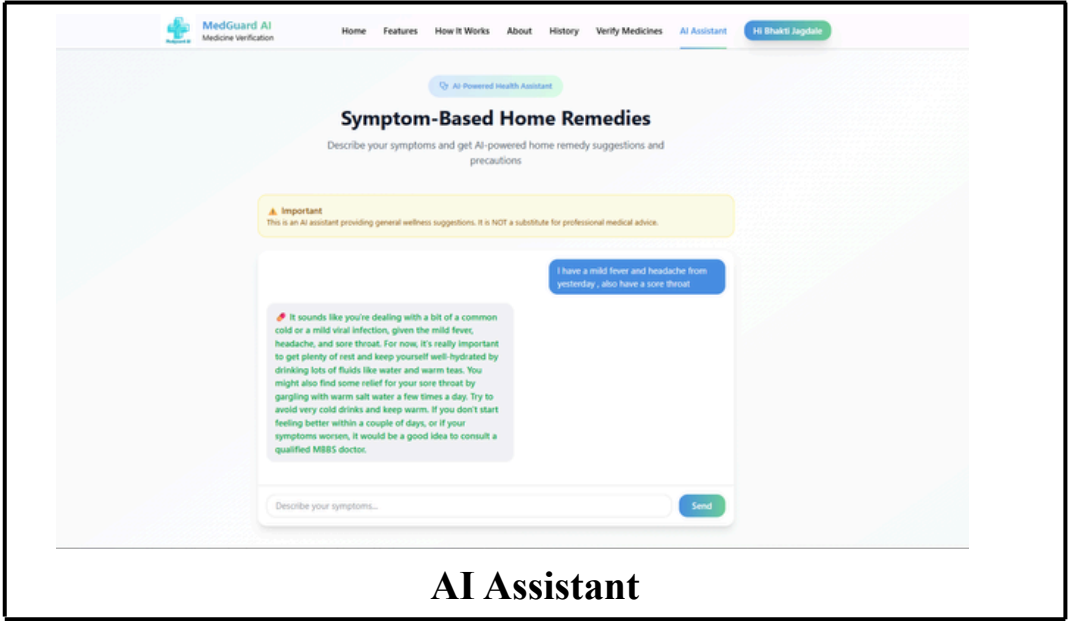
RESULTS



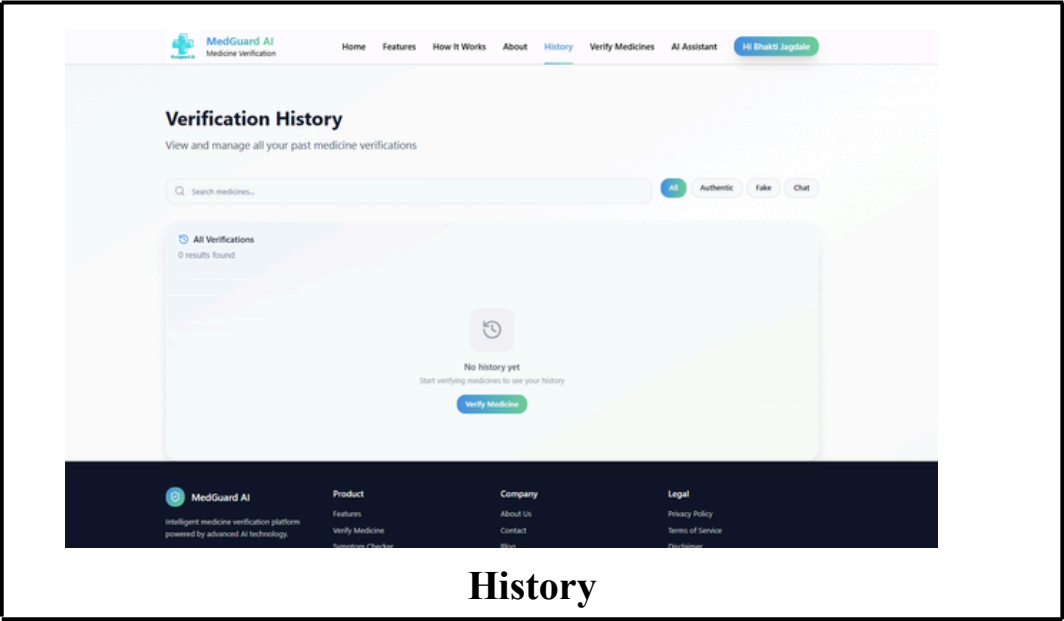
Home Page



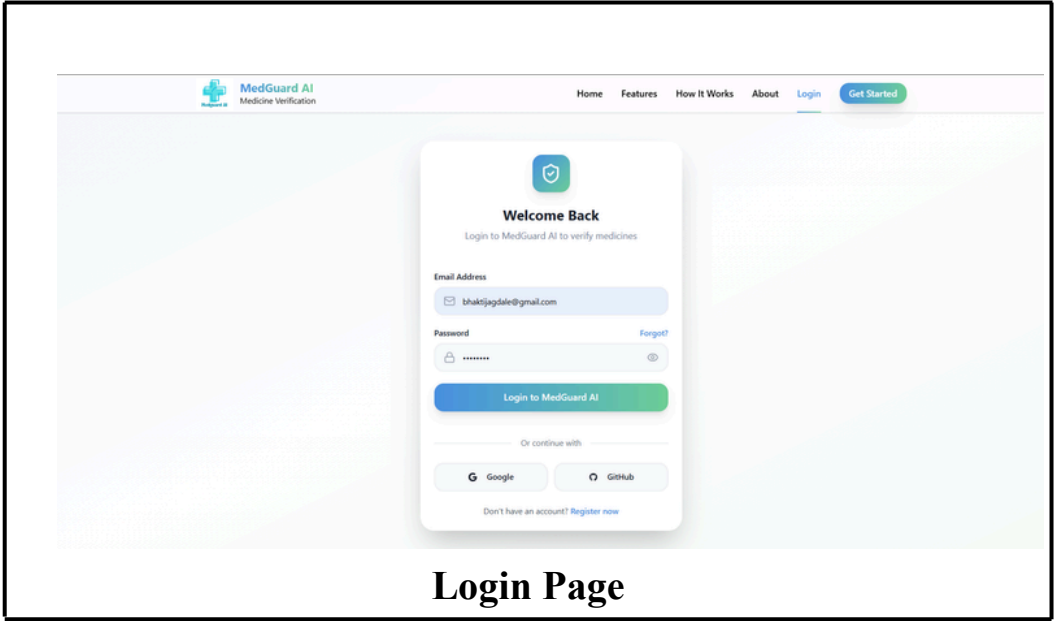
Verification Page



AI Assistant



History



Login Page



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CONCLUSION AND FUTURE WORK

Conclusion :

1. Successfully developed a multi-modal AI-based verification system
2. Improved reliability in medicine authentication
3. Provides accessible and practical solution for users

Future Work :

1. Expand dataset for higher accuracy
2. Improve AI model performance
3. Launch mobile application
4. Integrate real-time regulatory databases
5. Deploy fully on cloud for large-scale use



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